

Mathematical modeling and energy consumption evaluation of R-600a refrigeration systems for vending machines using efficient heat exchangers

Alina Girip^a, Răzvan Calotă^a, Ioan Mihail Savaniu^{b,*} , Anica Ilie^a, Emanuil Ovadiuc^a, Oana Tonciu^b

^a Technical University of Civil Buildings, Bucharest, Romania, Thermal Sciences Department

^b Technical University of Civil Buildings, Bucharest, Romania, Mechanical Engineering Department

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ABSTRACT

This study presents the SVIEE-R vending machine as a sustainable solution for reducing CO₂ emissions in the retail sale of cold packaged products, maintained at refrigeration temperatures between 5 °C and 8 °C. The system operates on a 12 V DC cooling unit, utilizing R-600a refrigerant, which has a low Global Warming Potential (GWP = 3) and zero Ozone Depletion Potential (ODP). Unlike conventional vending machines, the SVIEE-R is specifically designed for off-grid applications, integrating a flexible, dynamic, data-driven analytical tool for performance evaluation, prediction, and optimization. The novelty of this work lies in the combination of a natural refrigerant and an energy-efficient design tailored for autonomous operation in remote locations.

To enhance energy efficiency, an adaptive mathematical model is developed for the vapor compression refrigeration system of the vending machine. This model integrates thermodynamic principles, geometric parameters, and operational conditions to optimize performance. The key performance metrics, validated through experimental data with deviations below 10%, confirm the model's accuracy. The Coefficient of Performance (COP), determined using the Nusselt-Jürges correlation, ranged between 1.321 and 1.476, emphasizing the critical role of evaporator geometry and condensation temperature in system efficiency. This study provides a framework for adapting energy-efficient refrigeration technologies to diverse operational environments, thereby advancing the implementation of sustainable refrigerants in small-scale cooling applications.

1. Introduction

The reduction of greenhouse gas emissions in the retail sale of cold products can be achieved by using independent and energy-efficient vending machines such as the SVIEE-R vending machine. The SVIEE-R vending machine is an innovative solution allowing the storage and sale of cold products using a cooling system powered by 12 V DC. The SVIEE-R vending machine used in the experimental validation of the model meets the requirements of the European Green Deal [1] adopted by the European Union. According to this document, Member States need to achieve annual energy savings of 1.49% on average over the period 2024–2030.

Communities are moving towards healthier products by using vending machines that ensure lower prices for the products sold due to reduced personnel costs and the possibility of supplying new products during times when traffic is lower [2]. Energy-independent vending systems equipped with photovoltaic panels have several advantages,

including reduced electricity costs, flexible location, ease of installation, independence from the local electricity grid, environmental friendliness and reduced CO₂ emissions [3,4].

The study of energy efficiency for vending machines is particularly important for manufacturers and users of vending machines. It is regulated by the Energy Measurement Protocol EMP for refrigerated vending machine, developed by EVA European vending & coffee service association which establishes the need to mention the energy efficiency class, the mention of GWP of the refrigerant and recommends the use of energy-efficient equipment.

In recent years vending machines have become an important subject due to their importance in the food industry, in an era in which people tend to eat very fast and to have various options available all day long, without entering supermarkets. Several studies have been conducted in relation to energy efficiency and the best design, for improving the delivery of the products and services.

In Zhang et al. [5], the importance of propane (R-290) as an

* Corresponding author.

E-mail address: mihai.savaniu@utcb.ro (I.M. Savaniu).

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environmentally friendly refrigerant due to its low Global Warming Potential (GWP=3) is emphasized. The research highlights the flammability risks associated with R-290 leakage and introduces a Computational Fluid Dynamics (CFD) model to simulate and visualize flammable zones. By integrating a ventilation system with the refrigeration device, the flammability risks were mitigated, demonstrating the effectiveness of CFD in designing safer systems.

Manzano-Agugliaro et al. research [6] evaluates energy consumption in vending machines, identifying snacks, cold drinks, and hot drinks vending machines as consuming an average of 250 kWh, 200 kWh, and 100 kWh per month, respectively. Using a low-cost sensor network, the study successfully monitored energy usage and identified energy wastage patterns. It concluded that implementing such monitoring systems could significantly reduce the environmental impact and improve energy efficiency in public spaces.

Energy conservation strategies for refrigerated vending machines operating in high-temperature environments is explored in Wu et al. [7]. By introducing a dual-fan configuration to enhance airflow uniformity, energy consumption was reduced by 6.9%. Additionally, a heat recuperation mechanism achieved a 4.3% reduction in power consumption. These combined measures resulted in an overall energy saving of 11.2%, serving as a reference for optimizing vending machines in challenging climates.

Other works [8,9], investigate how visual cues on vending machine influence consumer behavior, underscoring the potential of subtle environmental nudges in shaping healthier consumer choices and highlighting the complex factors driving vending machine selections.

The study of energy-efficient refrigeration systems has primarily focused on optimizing refrigeration cycle performance, improving thermodynamic efficiency, and selecting low-GWP refrigerants. By integrating advanced modeling techniques and considering the practical aspects of vending machine operation in real-world environments, this research aims to optimize energy consumption and system performance to provide a framework for improving the energy performance of vending machines while also contributing to the objective of reducing greenhouse gas emissions.

To improve the current state of the art in regard to energy efficiency, in this study, the aim is to develop an automated calculation method for modeling the operation of vending machines utilizing R-600a refrigerant. The program is based on a steady-state vapor compression refrigeration cycle (VCRS) and calculates key performance metrics such as cooling capacity, compressor work, and the coefficient of performance (COP).

The research gap identified by the authors is the lack of an automated modeling approach for optimizing vending machine refrigeration systems using low-GWP refrigerants. While previous studies have explored energy efficiency and refrigerant safety, they have not provided a computational framework for real-time optimization for vending machines operating with R-600a.

This study addresses the gap by developing and validating an automated mathematical model for optimizing the steady-state vapor compression refrigeration cycle (VCRS).

R-600a, also known as isobutane, is a hydrocarbon refrigerant widely recommended for vending machines and household refrigerators due to its environmentally friendly properties. It has an Ozone Depletion Potential (ODP) of zero and a very low Global Warming Potential (GWP) of 3, making it a sustainable alternative to traditional refrigerants like R-134a, which have higher GWPs.

However, R-600a is classified as an A3 refrigerant by ASHRAE, indicating high flammability. This necessitates stringent safety measures, especially for indoor installations. Regulations often limit the refrigerant charge to minimize risks. For instance, in the United States, systems using R-600a are restricted to a maximum charge of 150 g.

In outdoor installations, such as the vending machine analyzed in this study, the flammability risk is significantly reduced due to better ventilation and the dispersion of any potential leaks. By integrating R-

600a into vending machines, manufacturers can achieve high energy efficiency and environmental sustainability, aligning with global efforts to reduce greenhouse gas emissions and to diminish ozone depletion.

The thermally powered solar vapor compression refrigeration system (VCRS) is designed and implemented in Bucharest, Romania. The site is located at 26.1° east longitude and 44.4° north latitude, in the middle of Ilfov County. In this region, the daily solar radiation during typical summer days (May-September) ranges from 800 to 998 W/m². In Bucharest, the average monthly ambient air temperature in August is approximately 30 °C, with a relative humidity of approx. 42%.

The performance of a photovoltaic (PV) panel system is influenced by three key meteorological parameters: ambient temperature, solar irradiation, and wind speed. Among these, solar irradiation has the most significant impact on the efficiency of the PV system, as it directly determines the electrical output. Studies have shown that in Bucharest, the average energy production per day for each kilowatt of installed solar capacity is approximately 6.70 kWh in summer, 3.25 kWh in autumn, 1.70 kWh in winter, and 5.12 kWh in spring [10].

Additionally, the efficiency of PV panels decreases with increasing ambient temperatures. For instance, a study found that for every 1 °C rise in temperature above 25 °C, the efficiency of a typical silicon PV cell decreases by about 0.5% [11].

Wind speed also plays a role in PV performance by affecting the cooling of the panels. Higher wind speeds can enhance convective heat loss from the panel surface, thereby reducing the negative impact of high ambient temperatures on efficiency. However, in urban areas like Bucharest, wind speeds are often lower due to building obstructions, which may limit this cooling effect.

Refrigerant cycle models are designed to optimize the VCRS by selecting the appropriate components for the evaporator, compressor, condenser, and the expansion valve.

Several studies have been conducted on modeling different types of refrigeration plants. In [12,13], CFD and numerical studies were employed to optimize centrifugal and scroll compressors, which parallels our study's focus on performance improvements in vending machine refrigeration components. In Mohtaram et al. [14], an experimental study was conducted to determine optimal conditions for electrical discharge machining, a methodology that, while distinct, provides insights into precision engineering, relevant to refrigeration system component design. Khan et al. [15] optimized the operation of an advanced cascaded recompression absorption system, a methodology that aligns with our study's approach to optimizing vending machine refrigeration systems for energy efficiency.

In Martínez-Ángeles et al. [16], the energy optimization of an auto-cascade refrigeration system operating with an R600a/R1150 mixture was investigated, focusing on compressor selection and capillary tube length adjustments to enhance system performance. Their study provides insights into the optimization of ultra-low temperature applications, particularly in small-scale systems where hydrocarbon refrigerants are used as alternatives to high-GWP fluids. Similarly, in Calleja-Anta et al., [17], a thermodynamic screening of alternative refrigerants for R290 and R600a was conducted, identifying potential refrigerant mixtures that could improve energy efficiency and reduce flammability risks in stand-alone refrigeration systems. This study represents the basis for selecting optimized refrigerants that can be further validated in experimental setups. These studies support the transition toward low-GWP refrigerants and optimized refrigeration cycles, aligning with the objectives of the present work.

In these basic models of refrigeration cycles, pressure drops are neglected, leading to the assumption of constant temperatures across the evaporator and condenser. While these simplifications aid initial analysis and system optimization, they limit the accuracy of the models under more dynamic or complex conditions especially for refrigeration systems with high cooling capacities. For more advanced modeling, the assumptions can be shaped to include the effects of pressure drops, heat losses, and non-uniform temperature distributions across the heat

exchangers. These advanced models enable better predictions of system performance under varying operating conditions, such as load changes and ambient temperature fluctuations. Moreover, they can incorporate the impact of refrigerant properties, flow characteristics, and component interactions, leading to more robust designs. In addition to performance optimization, these models are increasingly being used to evaluate environmental and economic aspects, such as refrigerant choice, energy consumption, and system life-cycle costs.

The purpose of this study is focused on an off-grid vending machine using R-600a refrigerant, powered by a photovoltaic system, and it aims to create an integrated and flexible model that can be used for design and performance predictions. The main objective of this work is to develop, validate, and apply a detailed mathematical model of an R-600a vending machine to analyze its energy performance under different conditions, with the aim of optimizing its design and efficiency and promoting the use of natural refrigerants. In order to achieve the objective, a mixed-methods approach combining experimental data collection from a real-world vending machine with the development and validation of a detailed thermodynamic mathematical model was used.

2. Experimental setup

The vending machine, illustrated in Fig. 1, is positioned at the Technical University of Civil Engineering Bucharest campus (UTCB), The SVIEE-R is located at latitude $44^{\circ}26'10.58''\text{N}$ longitude $26^{\circ}4'56.03''\text{E}$ with the solar panels facing South and with exposure from East to West as shown in Fig. 2. The equipment was designed to sell cold products packed in cylindrical aluminium or polyethylene terephthalate (PET) containers with a maximum capacity of 500 ml, diameter 65 mm and height 170 mm. An important feature of the SVIEE-R is that all component devices are powered by 12 V DC, this innovative solution reduces the risk of electrocution.

The design of the vending machine has considered the protection against vandalism. Thus, the acquisition of the products is done only by mobile payment through the dedicated app, where the availability of the products can be checked.



Fig. 2. SVIEE-R positioning in UTCB Campus.

The SVIEE-R automated system has the following subsystems: array of photovoltaic panels; batteries for solar energy storage; resistance structure; thermally insulated enclosure; cooling system; command and control system; delivery units.

Through the newly innovative technology for products dispensing, presented for the first time in this research study, the thermal energy losses are reduced, and the energy efficiency is higher than in conventional vending machines.

R-600a was chosen over other low-GWP refrigerants due to its very low GWP and zero ODP, making it one of the environmentally friendly options. Additionally, it offers high energy efficiency due to its thermodynamic properties, such as low boiling point, which contribute to reduced compressor work and lower energy consumption. Compared to alternatives like R-290 (propane) or HFO refrigerants, R-600a is well-suited for small-scale applications, such as vending machines and household refrigeration, where lower refrigerant charge and improved efficiency are important, despite its flammability requiring safety precautions.

The thermally insulated enclosure is a parallelepiped enclosure with interior dimensions 1270 mm x 850 mm x 750 mm, consisting of three

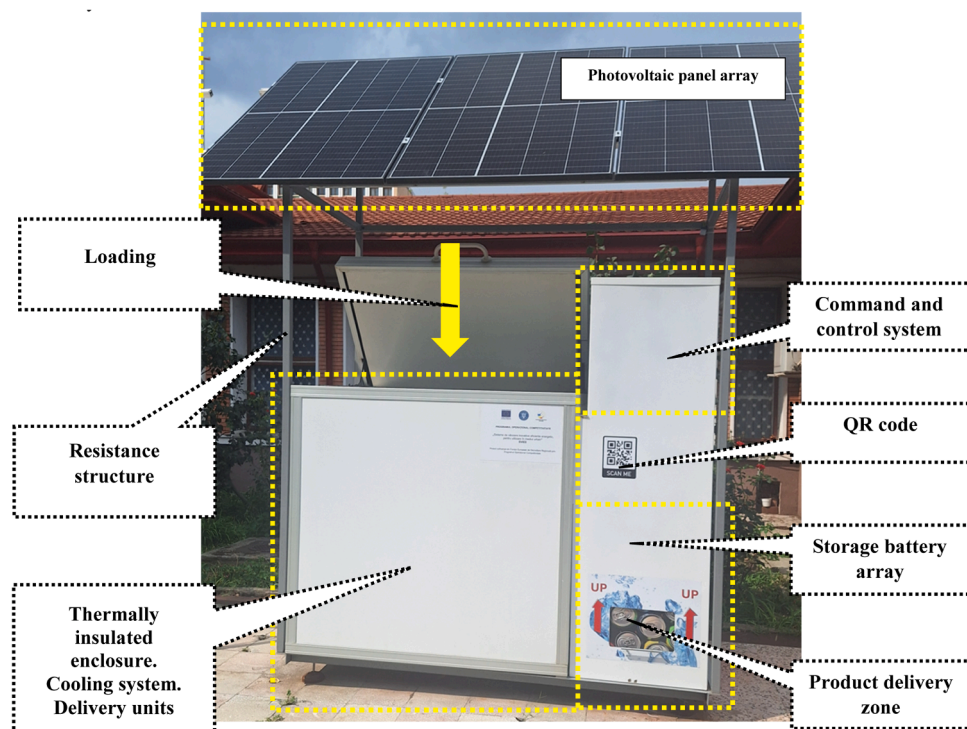


Fig. 1. SVIEE-R Vending Machine.

insulating layers: expanded PVC Komacel M2 ($\lambda = 0.05 \text{ W}/(\text{m}\cdot\text{K})$) 10 mm thick on the inside, Termoconfort M1 ($\lambda = 0.024 \text{ W}/(\text{m}\cdot\text{K})$) 100 mm thick in the middle and Bond PE M4 ($\lambda = 5.65 \text{ W}/(\text{m}\cdot\text{K})$) 3 mm thick on the outside. The products to be delivered are released by the mechatronic system of the delivery unit and then, gravitationally, due to the funnel shape of the lower area of the thermally insulated enclosure, they roll towards the delivery area.

The products are stored in delivery units and kept at a refrigeration temperature between 5°C and 8°C in a thermally insulated enclosure as shown in Fig. 3. The cold product commercialization process is as follows: the user selects the type of product from the stock of products available in the vending machine in the smartphone application; then, the purchase of the product is confirmed by scanning the QR code on the outer casing of the SVIEE-R; and the product is delivered by the system in the delivery area. Fig. 4

The development of the equipment considered the use, from the experimental model and prototype phases onward, of technologies that will enable the serial production of the vending machine at costs that make it economically viable in the cold-product vending market. The technology for fabricating the metal structure from standardized steel elements permits the use of mass-production methods at competitive costs. Similarly, the design of the thermally insulated storage enclosure -using a multilayer structure with a polyurethane foam core- enables the adoption of serial manufacturing techniques typical of the refrigerator industry. In addition, the use of catalog products for solar panels, batteries, and command and control equipment, along with standard catalog components for the refrigeration system, facilitates the series production of the SVIEE-R automatic machine at competitive costs. Moreover, the adoption of commonly available catalog equipment ensures quick, low-cost maintenance. A major advantage of developing a vending machine that uses refrigeration equipment powered by 12 V direct current is the absence of the inverter for converting the electricity stored in batteries into 230 V alternating current. By eliminating the inverter, the energy losses are reduced due to the conversion of direct current into alternating current, extending the autonomy of the vending machine. Inverters are expensive components, which raise the final cost of the vending machine, the research undertaken aimed at developing an automatic machine that uses usual, recyclable and safe components in operation.

For vending machines powered by photovoltaic panels, usage is limited by geographical location and the orientation relative to solar radiation. The SVIEE-R automatic machine has the advantage of operating without a connection to the electricity grid; however, it must be positioned so that the photovoltaic panels face the sun. Moreover, photovoltaic-powered vending machines are primarily used during the hotter periods of the year, when solar exposure is at its peak.

The components depicted in Fig. 5, illustrating the connection diagram of the energy-independent vending machine SVIEE-R, are the following:

1. **Solar panels:** Type TSM, 405 DE09.08, with a maximum power output of 405 W.
2. **MPPT controller:** Type Victron 150/70 Tr is responsible for managing energy flow from the photovoltaic array to the battery bank. Known for their high efficiency, MPPT controllers ensure optimal voltage conversion for battery charging, achieving a DC/DC conversion efficiency of 98%.
3. **Battery bank:** Type Victron GEL 12 V, 130 AH, providing energy storage for uninterrupted operation.
4. **Evaporator:** Type Guyan GNKF 850 YAN, equipped with two 12 V DC motors, each consuming 0.28 A.
5. **Condenser:** Dimensions $250 \text{ mm} \times 220 \text{ mm}$, equipped with two 12 V DC motors, each consuming 0.28 A.
6. **Command and control system:** Based on an ESP-WROOM-32D microcontroller, developed in-house.
7. **Compressor:** Hermetic brushless DC motor-compressor, model QDZY50G.
8. **Delivery units:** Powered by 12 V DC motors (model JGB 3530), also developed in-house.
9. **Power quality analyzer:** HIOKI PW3198, employed for monitoring electrical parameters during operation [18,19].

The experimental setup involved loading the vending machine's delivery units with 240 aluminum cans, each with a capacity of 500 ml, as shown in Fig. 6a. The cans were initially at ambient temperature before being cooled and maintained at refrigeration temperatures. During the testing period, the thermal conditions inside the insulated enclosure were monitored using three TESTO 184 H1 hydro-thermometers, strategically placed, as shown in Fig. 6b. An additional hydro-thermometer was used to monitor the outdoor temperature, as shown in Fig. 11. The recorded temperature and humidity are presented in Figs. 7–9.

The fluctuations in temperature depicted from Figs. 7 and 8 show the cyclical operation of the refrigeration system, as it works to maintain the interior at the desired level by periodically removing heat.

To monitor the electrical energy consumption of the compressors, a HIOKI PW3198 power quality analyzer was employed, positioned at location 9 in the schematic shown in Fig. 5 and integrated into the SVIEE-R. Fig. 10 ensures detailed visualization. The analyzer was powered by an external 230 V AC source and featured high measurement accuracy, with voltage precision of $\pm 0.1\%$ of the nominal value, current accuracy of $\pm 0.2\%$ rdg. $\pm 0.1\%$ f.s. combined with current sensor



Fig. 3. Products' delivery steps.

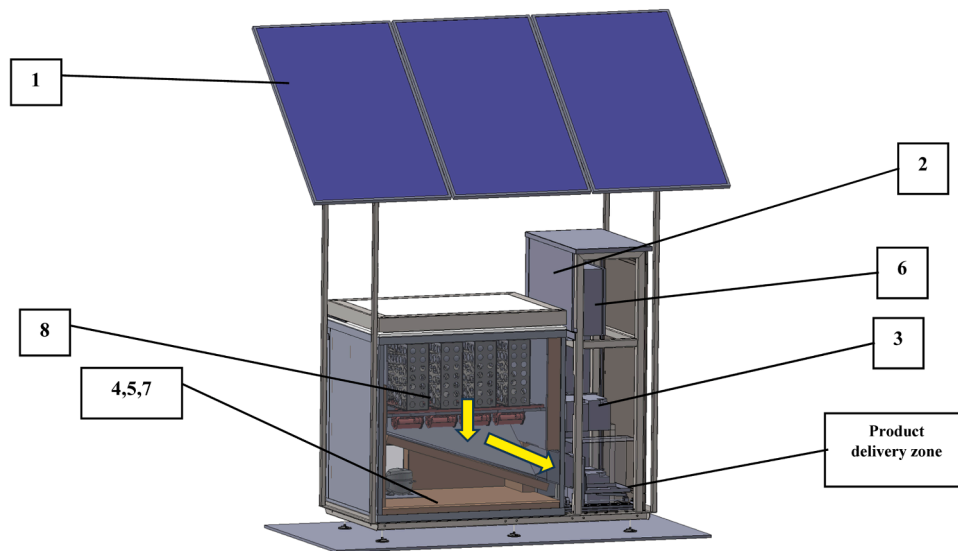


Fig. 4. SVIEE-R Components.

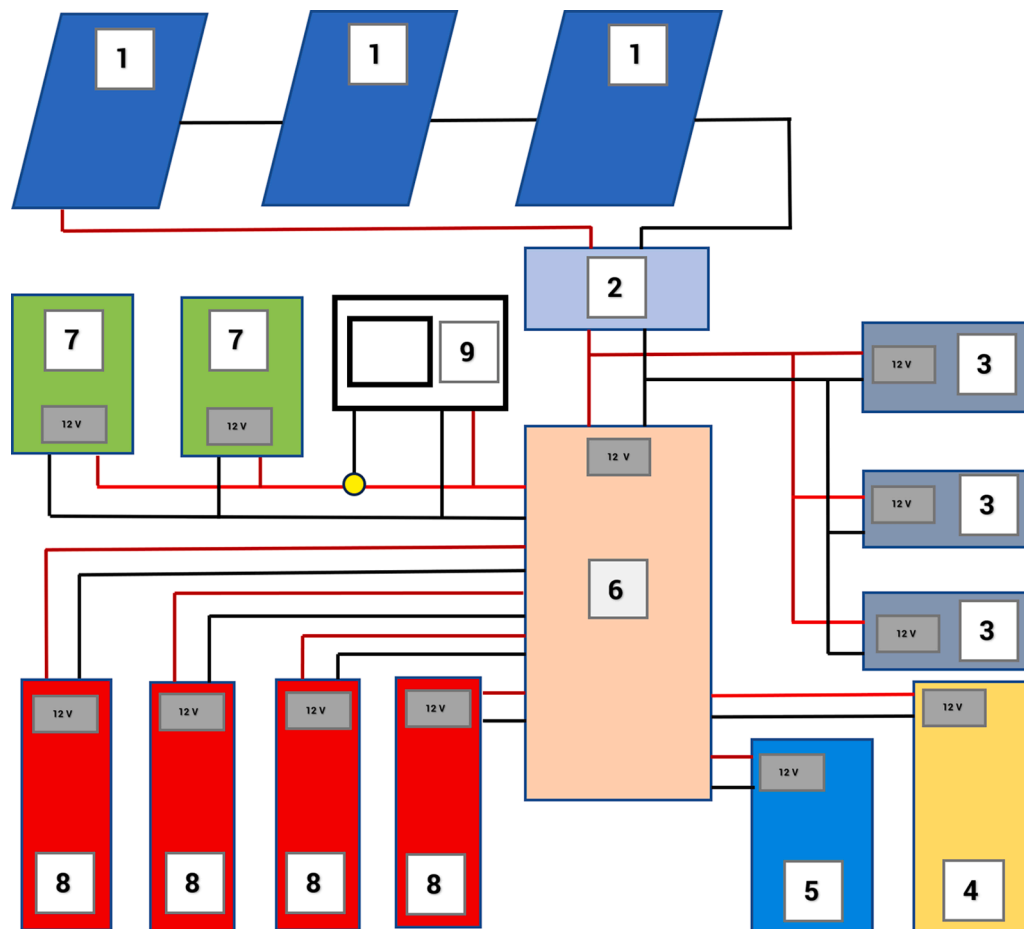


Fig. 5. SVIEE-R Connecting scheme.

accuracy, and active power accuracy of $\pm 0.2\%$ rdg. $\pm 0.1\%$ f.s. plus current sensor precision during the testing period. The HIOKI PW3198 continuously recorded data on supply voltage, current, and the electrical energy consumed by the system. The collected data were stored in the analyzer's internal memory and subsequently analyzed using PQ ONE software (version 10.2.0). This fact ensured detailed visualization of

compressor energy consumption in both graphical and tabular formats.

Additionally, electricity production from the photovoltaic panels was monitored during the test period. Data from the Maximum Power Point Tracker (MPPT) controller were collected using the Victron Energy VE.Direct to USB interface, as depicted in Fig. 11. The tests commenced with the batteries fully discharged, and the Programmable Logic



Fig. 6. a) The method of loading the SVIES with doses; b) The sensors placement.

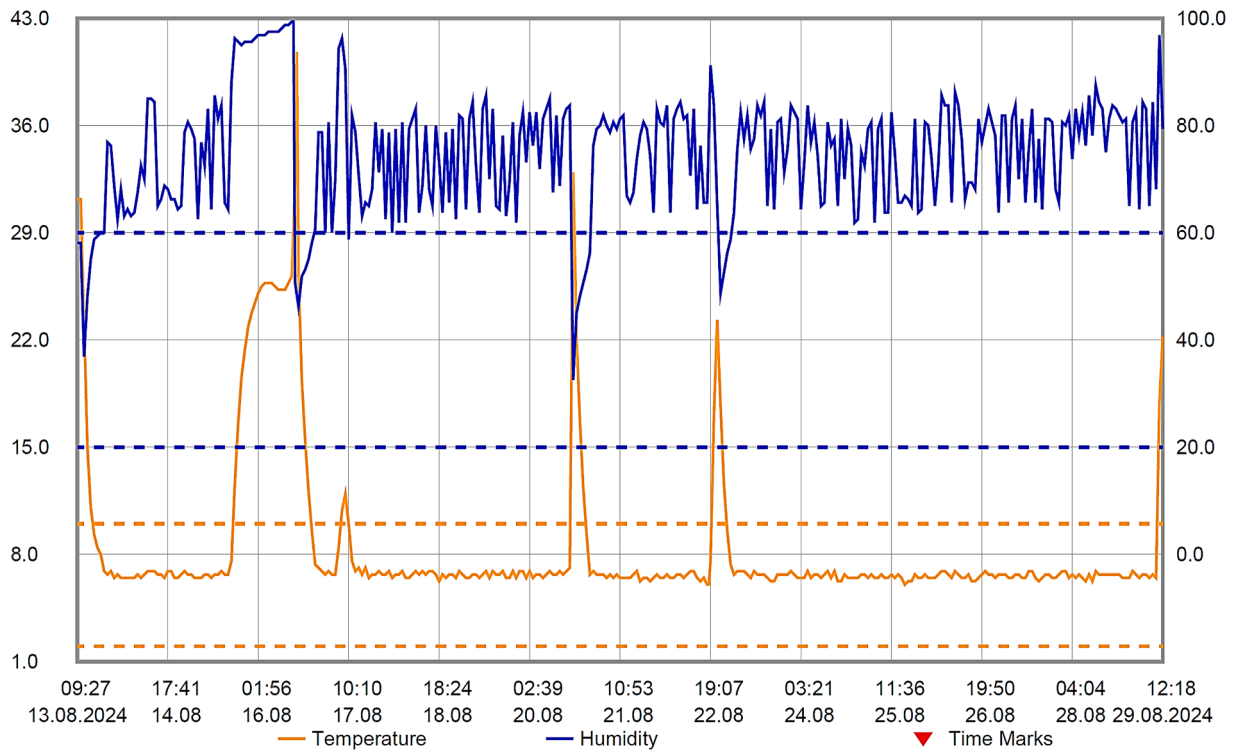


Fig. 7. The temperature inside the Vending Machine recorded by the SI sensor.

Controller (PLC) turned off to establish a baseline prior to initiating energy monitoring. The results, including the total converted energy (Wh) and the maximum power output of the PV system during the test period, are presented in Fig. 12.

The relative uncertainty of the direct measurement is as follows: temperature 2.32%, pressure 1.81%, and total power consumption of the vending machine 0.71%.

The manufacturing processes for the SVIEE-R consider environmental sustainability both during production and at the end of its life. For example, the metal structure is made entirely of recyclable steel, and the thermally insulated tank is produced using standard refrigerator manufacturing techniques -comprising an outer layer of steel, a middle layer of polyurethane foam for insulation, and an inner layer of PVC-with existing recycling technologies available for these materials. Additionally, the vending machine's components -including photovoltaic panels, batteries, the command-and-control system, and refrigeration units- are fully recyclable at specialized centers. Furthermore, all components were selected with RoHS (Restriction of Hazardous Substances Directive) certification in mind

3. Mathematical model

Table 1 shows the input required for the mathematical model.

The isentropic efficiency of the compressor is calculated using a cross-term correlation, depending on condenser and evaporator temperatures. The outputs from the mathematical model are presented in Table 2.

3.1. Mathematical model description

The VCRS Cycle is presented in Fig. 13. To develop an automated calculation method for modeling the operation of vending machines utilizing R-600a refrigerant, polynomial equations were derived based on existing tabulated data for thermodynamic properties at saturation, as well as for the superheated states. This represents an original contribution of the authors.

In order to identify every point of the thermodynamic cycle, their position is specified in Table 3. The diagram used as a model is imported from Ashrae Handbook [20].

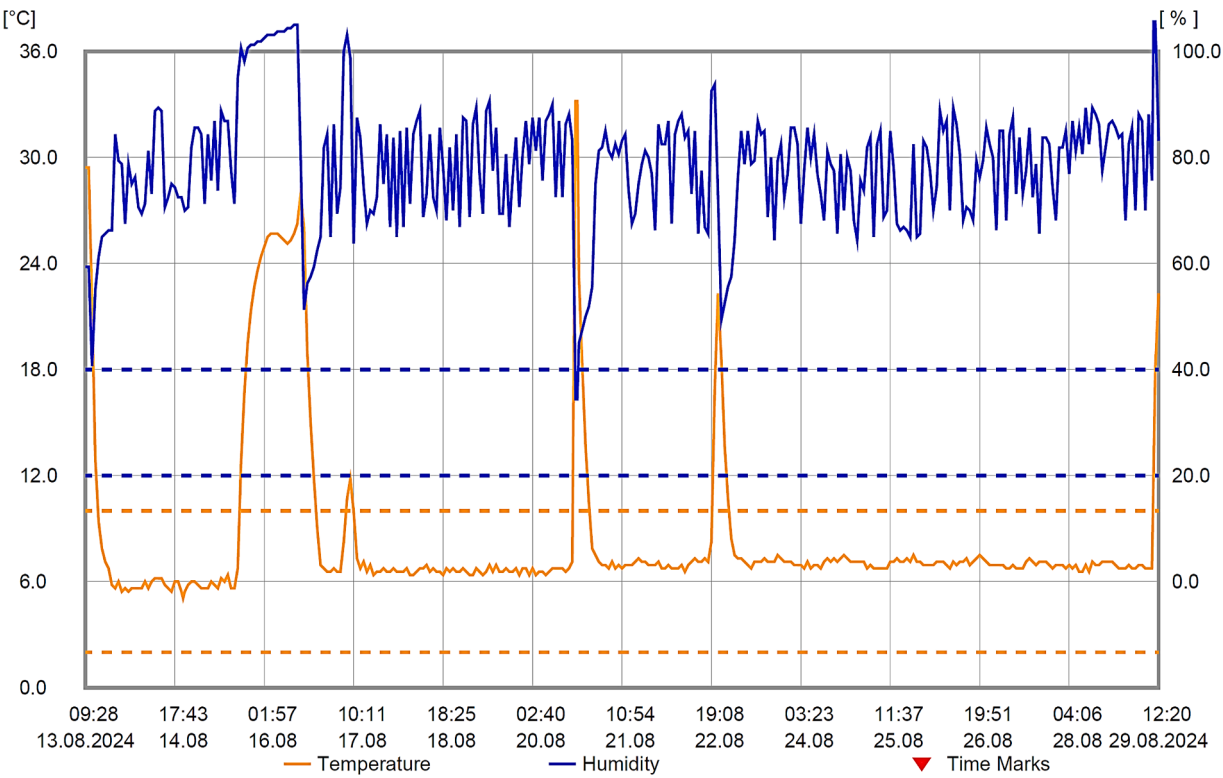


Fig. 8. The temperature inside the Vending Machine recorded by the SII sensor.

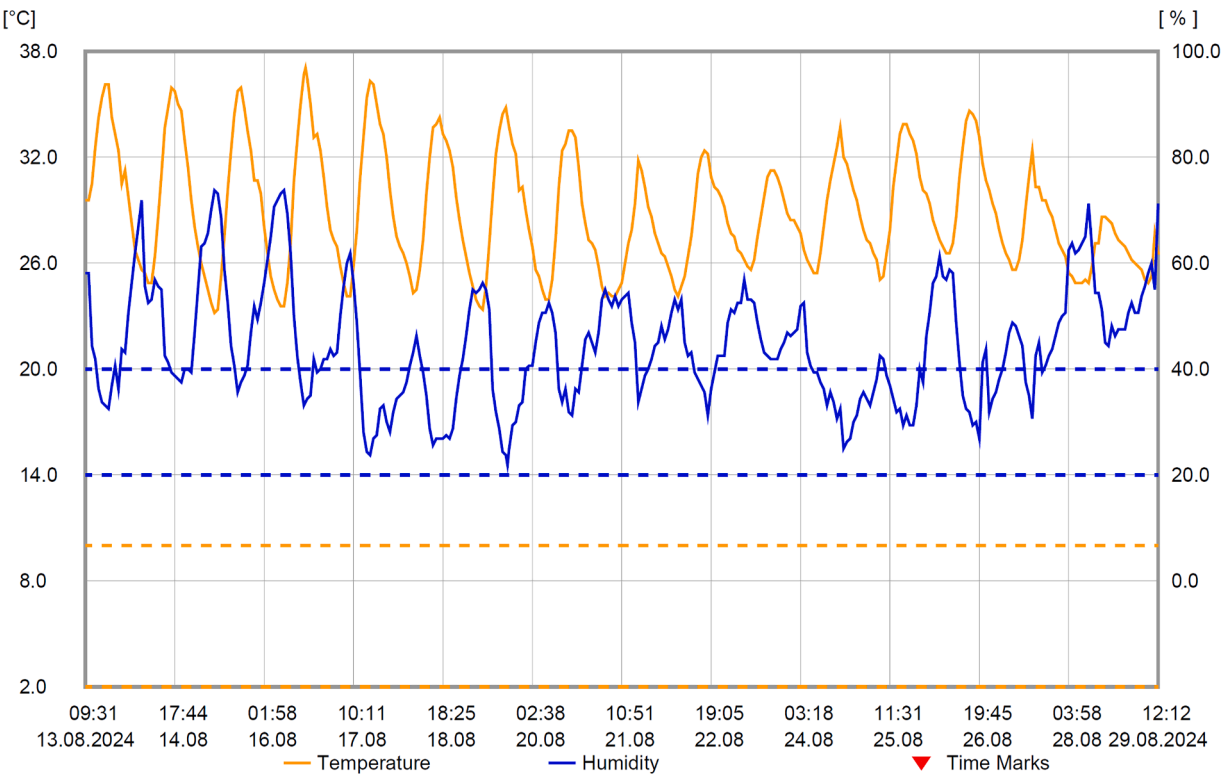


Fig. 9. The temperature at the Vending Machines exterior.

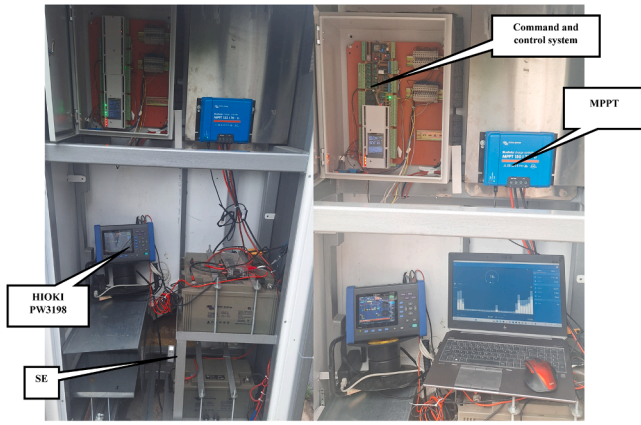


Fig. 10. Installation of the HIOKI PW3198 current analyzer and monitoring of the photovoltaic system.

Eqs. (1–5) are followed by Table 4 in which the maximum [%] deviation is presented.

$$p_{sat}(t_{sat}) = -1.1918 \cdot 10^{-10} \cdot t^5 - 3.4649 \cdot 10^{-10} \cdot t^4 + 4.3699 \cdot 10^{-6} \cdot t^3 + 7.566 \cdot 10^{-4} \cdot t^2 + 5.5638 \cdot 10^{-2} \cdot t + 1.5697 \text{ [bar]} \quad (1)$$

$$h_1(t, p) = 4867.61 + 525.33 \cdot t - 6715.44 \cdot p + 13.71 \cdot t^2 - 382.56 \cdot t \cdot p + 2534.7 \cdot p^2 \text{ [kJ/kg]} \quad (2)$$

$$h_3(t, p) = 255.02 + 3.55 \cdot t - 31.41 \cdot p - 0.04 \cdot t^2 + 1.21 \cdot t \cdot p - 4.5 \cdot p^2 \text{ [kJ/kg]} \quad (3)$$

$$s_1(t, p) = 2.27699 + 0.03916 \cdot p - 0.002266 \cdot t - 3.5 \cdot 10^{-5} \cdot t^2 + 0.001242 \cdot t \cdot p - 0.009758 \cdot p^2 \text{ [kJ/kg·K]} \quad (4)$$

$$v_1(t, p) = 0.36377 + 3.3 \cdot 10^{-5} \cdot p - 0.009976 \cdot t - 0.052289 \cdot t^2 + 0.00673 \cdot t \cdot p - 3.5 \cdot 10^{-5} \cdot p^2 \text{ [m}^3/\text{kg]} \quad (5)$$

Eqs. (1–5) provide a specific, customized, and computationally

efficient way to handle thermodynamic properties within this study's requirements. This contributes to the novelty and practicality of the study because the authors are generating a new tool that could be easily used by other researchers, without requiring the specific use of computational software, that is sometimes hard to correlate.

In isentropic compression, the entropy remains constant throughout the entire process. The refrigerant state at the discharge is superheated vapor, and at the suction is also superheated vapor. Polytropic compression is calculated by incorporating the isentropic compression efficiency:

$$h_2 = h_1 + \frac{h_{2is} - h_1}{\eta_{is}} \text{ [kJ/kg]} \quad (6)$$

The isentropic compression efficiency is calculated using a correlation of the following type:

$$\eta_{is} = T_0/T_c[-] \quad (7)$$

For the capillary tube rolling process, it is considered that the process takes place at constant enthalpy $h_3 = h_4$.

In the automated calculation method, the evaporator heat load is obtained by adding the infiltration load through the vending machine's walls $\dot{Q}_1$, heat load from evaporator fans $\dot{Q}_{fan}$ and from products introduction $\dot{Q}_3$:

$$\dot{Q}_1 = \frac{t_e - t_i}{R} S_{lat} \text{ [W]} \quad (8)$$

$$\dot{Q}_3 = \frac{\dot{m}_{prod} \cdot c_{prod} \cdot (t_{prod} - t_i)}{\tau_{prod}} \text{ [W]} \quad (9)$$

$$\dot{Q}_{TOT} = \dot{Q}_1 + \dot{Q}_3 + \dot{Q}_{fan} \text{ [W]} \quad (10)$$

Where:

R – total thermal resistance of the vending machine envelope [$\text{m}^2\text{K/W}$] calculated as:

$$R = \frac{1}{h_{int}} + \sum \frac{\delta}{\lambda} + \frac{1}{h_{ext}} \text{ [m}^2\text{K/W]} \quad (11)$$

S_{lat} – vending machine's lateral surface [m^2].

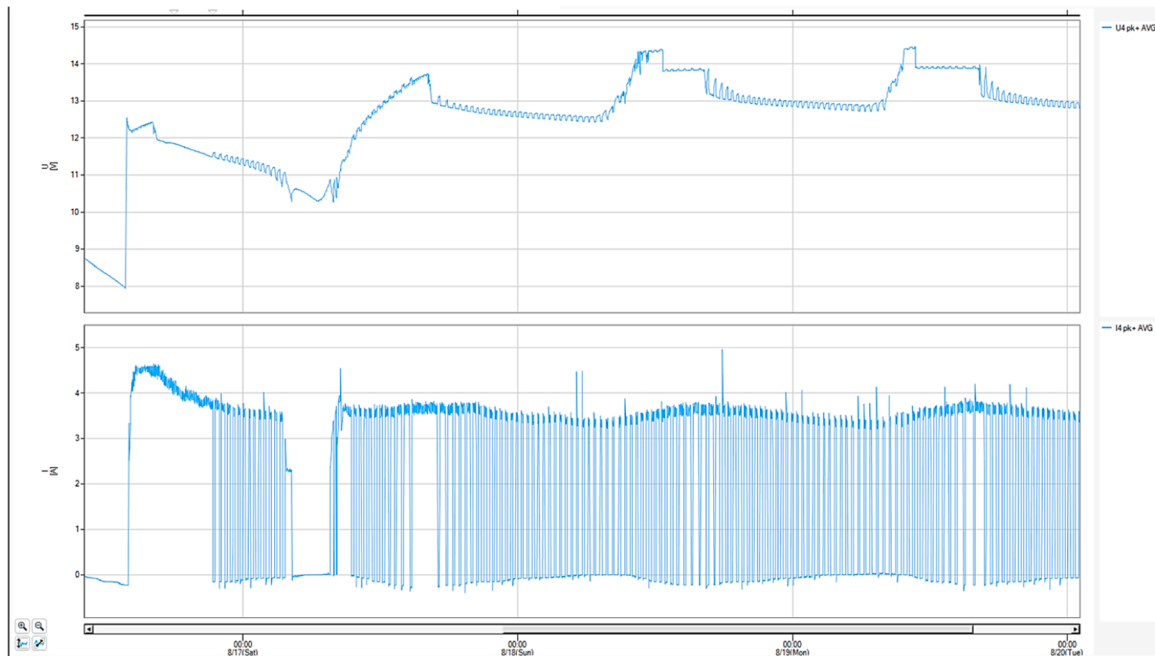


Fig. 11. Voltage and current recorded during the test period by the HIOKI PW3198.

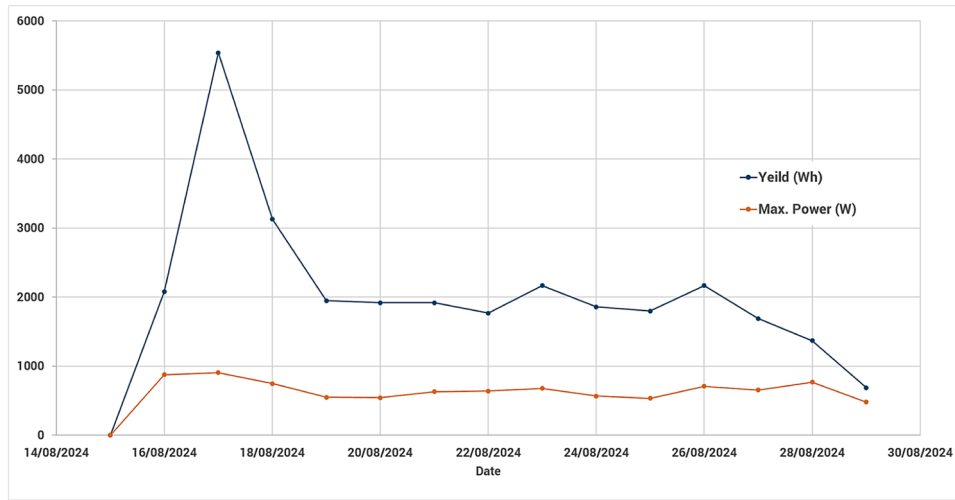


Fig. 12. Energy production and maximum power of photovoltaic panels during the test period.

Table 1

Inputs/parameters of the model.

Input parameters	Values
Set point temperature inside vending machine [°C]	$t_i = 6...8^{\circ}\text{C}$
Evaporation temperature[°C]	t_0
Temperature outside the vending machine [°C]	t_e
Air velocity outside the vending machine [m/s]	w
Condenser temperature [°C]	t_c
Evaporator (E) outlet temperature [°C]	$t_{out,E} = t_1$
Condenser (C) outlet temperature [°C]	$t_{out,C} = t_3$
Conductive thermal resistance [m ² K/W]	4.44
Products mass at the vending machines interior (240 cans of 0.36 kg each)	$m_{prod.} = 86.4 \text{ kg}$
Specific heat of products [J/kgK]	$c_{prod.} = 4188$

The total conductive thermal resistance, as expressed in Eq. (11), comprises three distinct layers: an outer layer of expanded PVC Komacel M2 ($\lambda = 0.05 \text{ W}/(\text{m}\cdot\text{K})$) with a thickness of 10 mm, a central layer of Termoconfort M1 ($\lambda = 0.024 \text{ W}/(\text{m}\cdot\text{K})$) measuring 100 mm in thickness, and an inner layer of Bond PE M4 ($\lambda = 5.65 \text{ W}/(\text{m}\cdot\text{K})$) with a thickness of 3 mm.

The term $\dot{Q}_3$ refers to the total product cooling load required to cool the products from the storage temperature $t_{prod.}$ to the set temperature from the vending machines interior t_i .

3.2. Heat transfer on the air side

To execute the mathematical model, input data were obtained from sensors, thermos-physical properties derived from data tables or empirical equations developed by the authors, and geometrical information sourced from the vending machine design schematics. The

Table 2

Outputs from the mathematical model.

Outputs
Evaporator pressure [°C] – p_0
Condenser pressure [°C] – p_c
Mass flow rate of the refrigerant [kg/s] – $\dot{m}_{R600a}$
Heat rejected by the condenser [kW] – $\dot{Q}_c$
Cooling capacity [kW] – $\dot{Q}_0$
Electrical power of the system [kW] – P_k^{th}
Coefficient of performance [-] – $COP_{cooling}$
Global coefficient of performance [-] – $COP_{global \ cooling}$
Isentropic compression efficiency [-] – η_{is}

calculation of the thermal resistance of the vending machine wall requires not only geometrical data and material properties to determine conductive resistance but also data on air circulation to estimate the convective resistances on both the interior and exterior surfaces.

To achieve this, several methods for estimating convective heat transfer coefficients are available in the literature. For this study, the authors selected five empirical equations to estimate the exterior convective heat transfer coefficients, ensuring a robust and comparative approach to assess the convective resistances.

Over the years, experimental correlation models of the convective heat transfer coefficients (CHTCs) have been developed. These models are generally classified into two main approaches:

1. Power Equation Model:

The first approach involves the use of the Nusselt number in combination with the Grashof and Prandtl numbers, expressed in the form of a power equation:

$$Nu = C \cdot (Gr \cdot Pr)^n \quad (12)$$

Where:

C , n are constant, experimentally determined as a function of the type of the flow and the surface shape.

2. Simplified Temperature-Based Equation:

The second approach uses a simplified equation of the form:

$$h = C \cdot (\Delta t)^n \quad (13)$$

Where Δt represents the temperature difference between the average temperatures of the plate and air.

The surface heat transfer coefficients for the wall, floor, and ceiling were calculated based on measurements of ambient temperature (t_e) and surface temperature (t_s) under natural convection conditions. The empirical formulas used are provided in Tables 5.1 and 5.2. These values typically depend on the characteristic dimension, the temperature difference between the environment and the surface, and airflow characteristics (air velocity - w).

Due to the variability in the obtained data -only McAdams and Nusselt Jurges correlation give similar values- the authors have opted to continue the study using the full dataset. During the validation phase, theoretical data will be compared with monitored data, and conclusions regarding the correlation with the minimum error will be drawn.

Fig. 14 illustrates that the convective heat transfer coefficient on the exterior surface exhibits higher values in correlations that account for air velocity. This observation aligns with expected trends, highlighting the differences between natural convection and forced convection.

The heat load of the evaporator fan motors is dependent on their

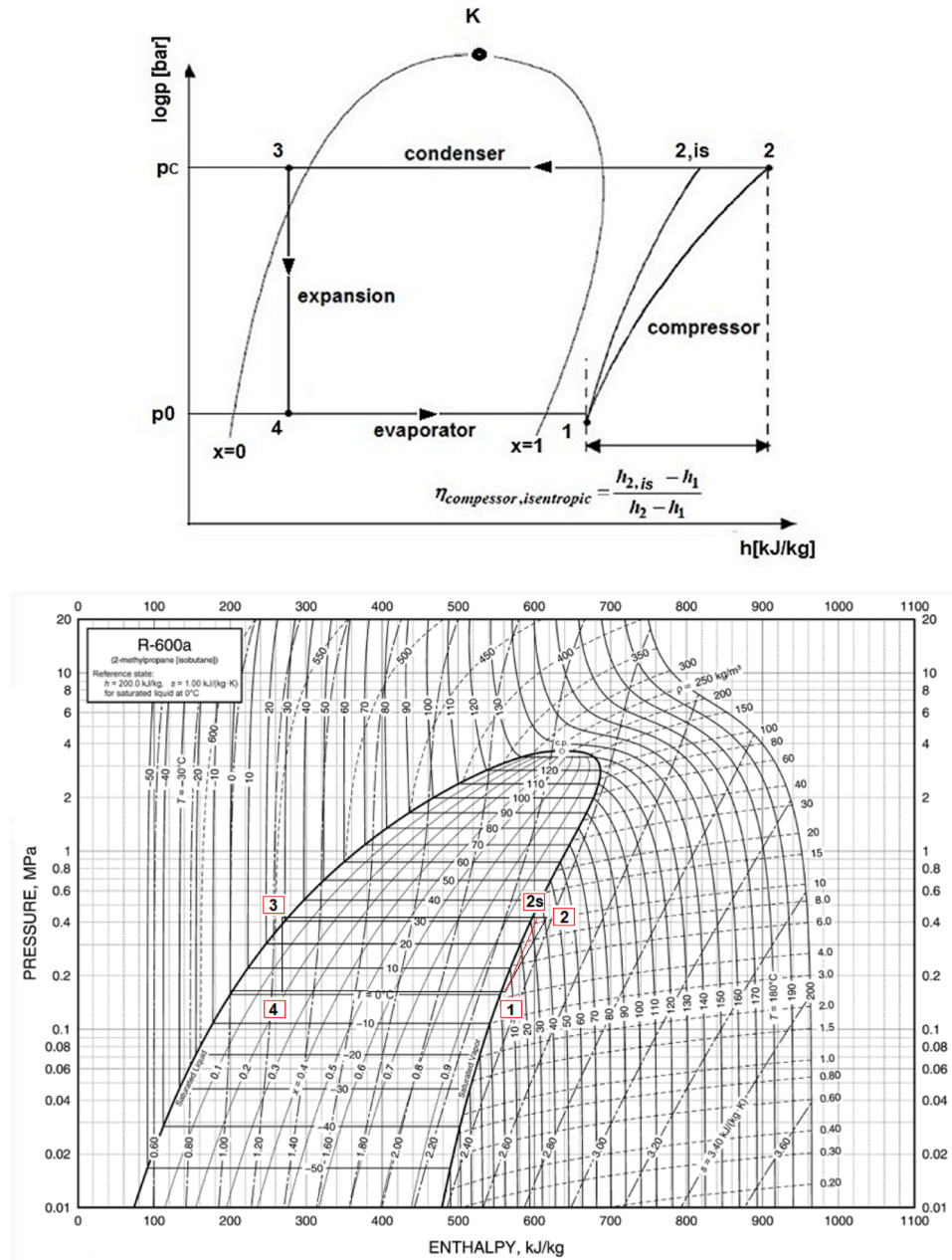


Fig. 13. Thermodynamic cycle a). Theoretical cycle; b).R600a cycle with thermodynamic states (current study).

Table 3

Thermodynamic states.

Calculated points	Description (position)
1	Compressor's inlet
2 _{is}	End of the isentropic compression (ideal)
2	End of the polytropic compression (real)
3	Condenser's outlet (after subcooling)
4	Evaporator's inlet

Table 4

Deviation in [%] for the determined equations.

Equation	Maximum deviation from the measuring interval [%]
(1)	0.11
(2)	0.42
(3)	0.53
(4)	0.14
(5)	0.63

efficiency given by the Eq. (14).

$$\dot{Q}_{fan} = P_{el.} / \eta \quad (14)$$

Where:

$P_{el.}$ – fan power [W];

η – fan motor efficiency [-].

For each component of the VCRS, the mathematical equations used to obtain the energy balance are presented in the next paragraphs. The refrigerant mass flow rate is calculated as:

$$\dot{m}_{R600a} = \frac{\dot{Q}_{TOR}}{h_1 - h_4} [kg / s] \quad (15)$$

The condenser rejected heat is obtained from Eq. (16):

Table 5.1

Correlations used to estimate convective heat transfer coefficient at exterior side.

Exterior surface	Correlations	Range	Heat transfer type
Vertical surface (Ashrae) [20]	$h_{ext} = 1.33 \cdot \left(\frac{t_e - t_s}{L} \right)^{1/4}$	$10^5 < Ra < 10^9$	Natural convection in free space
	L – hydraulic diameter		
Horizontal surface (Ashrae) [20]	$h_{ext} = 1.26 \cdot (t_e - t_s)^{1/3}$	$Ra > 10^9$	
	L – hydraulic diameter		
Nusselt [21]	$h_{ext} = 1.04 \cdot \left(\frac{t_e - t_s}{L} \right)^{1/4}$	$10^5 < Ra < 10^9$	Forced convection
	L – hydraulic diameter		
McAdams [22]	$h_{ext} = 5.7 + 3.8 \cdot w$	–	Forced convection
Mitchell [23]	$h_{ext} = 8.6 \cdot w_{air}^{0.6} / A^{0.4}$ A – is the cube root of the vending machine's total volume	–	Forced convection
M.A. Miheev [24]	$h_{ext} = 0.54 \cdot Ra^{0.25} \cdot k/L$ L = for plane plate low dimensions	$Ra < 500 \dots 2 \cdot 10^7$	Natural convection in free space
	$h_{ext} = 0.135 \cdot Ra^{0.33} \cdot k/L$ L = for plane plate low dimensions	$Ra > 2 \cdot 10^7$	

Table 5.2

Correlation used to estimate convective heat transfer coefficient at interior side.

Interior surface	Correlation	Range	Mode of heat transfer
Pohlhausen [25]	$h_{int} = 0.664 \cdot Re^{0.5} \cdot Pr^{0.33} \cdot k/L$ L – plate dimension in flow direction	$0.6 < Pr < 2000$ Laminar $Re < 10^5$	Forced convection

$$\dot{Q}_c = \dot{m}_{R600a} \cdot (h_2 - h_3) [W] \quad (16)$$

In order to calculate the compressor work, the following equation is implemented:

$$P_k^{th} = \dot{m}_{R600a} \cdot (h_2 - h_1) [W] \quad (17)$$

The reciprocating model of the compressor gives the external pressure ratio RC defined as in eq. (18):

$$RC = \frac{P_c}{P_0} [-] \quad (18)$$

The coefficient of performance in cooling mode for the VCRS is:

$$COP_{cooling} = \frac{\dot{Q}_{TOT}}{P_k^{th}} [-] \quad (19)$$

The global coefficient of performance (20) in cooling mode of the VCRS is calculated as:

$$COP_{global\ cooling} = \frac{\dot{Q}_{TOT}}{P_{Total}} = \frac{\dot{Q}_{TOT}}{P_k^{th} + P_{fan\ evaporator} + P_{fan\ condenser}} [-] \quad (20)$$

The $COP_{global\ cooling}$ includes the electricity consumption of the compressor, evaporator and the condenser's fans without internal controllers.

Starting from the input data, thermal balance and necessary output, a flow chart was developed in order to highlight the entire workflow. This automated calculation model that can be inserted in any type of software like Mathcad, or Microsoft Excel can be used to evaluate the vending machine's refrigeration system performance, when the input parameters are varied. The workflow is presented in Fig. 15.

The total power consumption ($P_k^{th} + P_{fan\ evaporator} + P_{fan\ condenser}$) is calculated by using the above-presented mathematical model and is compared with the monitored total power consumption which was monitored by measuring sensors. The admitted tolerance for this comparison is a maximum of 4%.

4. Experimental setup

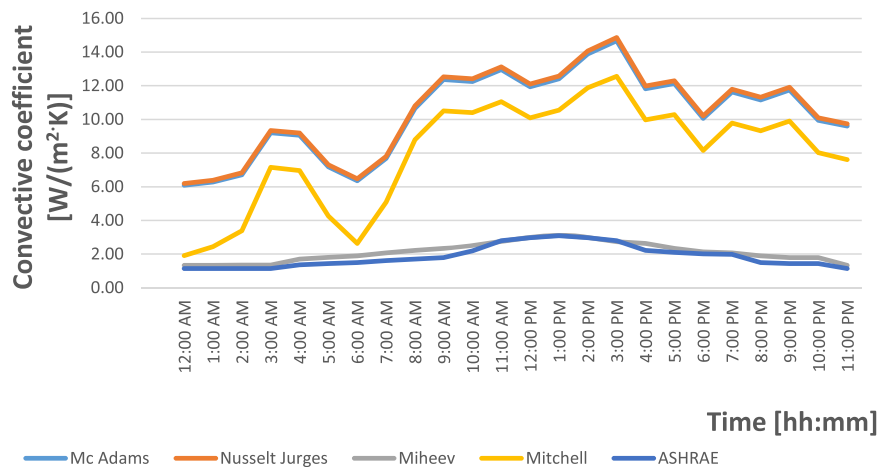
The evaluation of power consumption is strongly linked to the type of reciprocating compressor, the evaporator and condenser fans, and the thermodynamic properties of the refrigerant.

The technical data of each component from VCRS is presented in Table 6. Details on refrigeration unit components can be found in Fig. 15.

Since the vending machine operates off-grid and relies on solar radiation, outdoor weather conditions- such as temperature and solar radiation intensity- must be continuously monitored during its operation. The accuracy of the measuring instruments is detailed in Table 7.

Statistical measures can be used to perform an error analysis of the total power consumption of the refrigeration system by comparing the presented mathematical model against experimental data. This paper presents the average absolute percent error (AAPE) calculated for consecutive measurements in a given dataset. Fig. 16

$$AAPE = \frac{1}{m} \cdot \sum_m \left| \frac{P_k^{th} - P_k^{exp}}{P_k^{th}} \right| \times 100\% \quad (21)$$

**Fig. 14.** Exterior Convective Coefficient vs. Time.

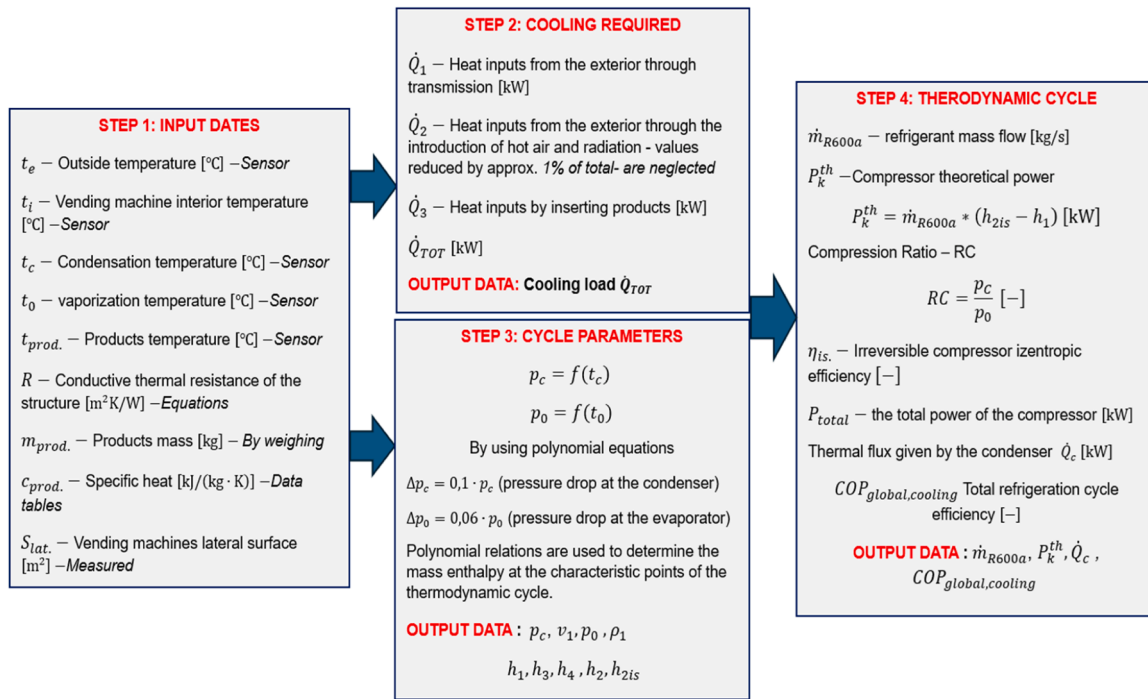


Fig. 15. Mathematical model workflow.

5. Results and discussions

The validation of the mathematical model is performed by comparing the compressor work calculated by the proposed mathematical model with measured data. The graph presented in Fig. 17 provides insights into the comparison between the monitored compressor power values and the theoretical values calculated using the five correlations outlined in Table 5.1.

Isentropic efficiencies are obtained from experimental data. The decreasing isentropic efficiency with the increasing pressure ratio, presented in Fig. 17, is due to the real compressor's inherent inefficiencies, including friction and internal heat generation that increase at higher compression ratios.

Based on Fig. 17, the polynomial fit Eq. (22), a result of the regression analysis, has a coefficient of determination R^2 of 99.98% as a function of pressure ratio:

Table 6
Equipment Technical Data Type.

Equipment	Technical data	Type
Evaporator coil	Length $L = 40$ cm Width $W = 90$ cm Height $H = 50$ cm Inlet/outlet connection 3/8"	GUNAY model GNF 850 YAN
Evaporator fans	Number of fans = 2 Model: 120 × 120 × 25 mm, 12 V DC, 0.28A, electrical power 6.72 W	ROHS model CFM A225BG-130-431
Condenser coil	Length $L = 28.2$ cm Width $W = 40$ cm Height $H = 22.5$ cm tube outer diameter 9.52 mm, number of coils: 8	Manufactured on command
Condenser fan motor	Model: 120 × 120 × 25 mm, 12 V DC, 0.28A, electrical power 3.96 W	Tidar
Hermetical reciprocating compressor	Displacement 5 cm ³ Oil charge 130 ml, power supply 12/24 V DC	SIKELAN QDZY 50G

Table 7
Measuring instruments.

Equipment	Measured Value	Accuracy	Measured Interval
Pyranometer	Intensity of solar radiation	±3%	0...1500 W/m ²
Wind velocity sensor Type FVA615-2	Air velocity	±3%	horizontal wind speed within a range of 0.5 to 50 m/s
USB type humidity sensor	Temperature (air)	±0.5 °C	−20...70 °C
USB type HIOKI PW 3198	Relative humidity	±3%	0...100%
3-Phase 4-Wire power quality analyzer	Single-phase 2-wire, Single-phase 3-wire, Three-phase 3-wire or Three-phase 4	Voltage: ±0.1% of nominal voltage Current: ±0.2 % rdg. ±0.1 % f.s. + current sensor accuracy Active power: ±0.2 % rdg. ±0.1 % f.s. + current sensor accuracy ±0.5 °C	Voltage: 600.00 V rms Current 500.00 mA to 5.0000 kA AC Power: 300.00 W to 3.0000 MW
TESTO 550 Digital Manifold	Evaporating and condensing temperature of R600a Evaporation and condensing pressure of R600a	0.01 bar	−50 to +150 °C −1 to 60 bar

$$\eta_{is} = 0.0065 \cdot RC^2 - 0.0764 \cdot RC + 1.0536 \quad (22)$$

Fig. 18 presents a detailed comparison of total energy consumption, showcasing the differences between experimental data and theoretical calculations.

The graph illustrates the total compressor power consumption over a 24-hour period, comparing directly monitored experimental data with theoretical predictions based on various correlations (McAdams, Nusselt

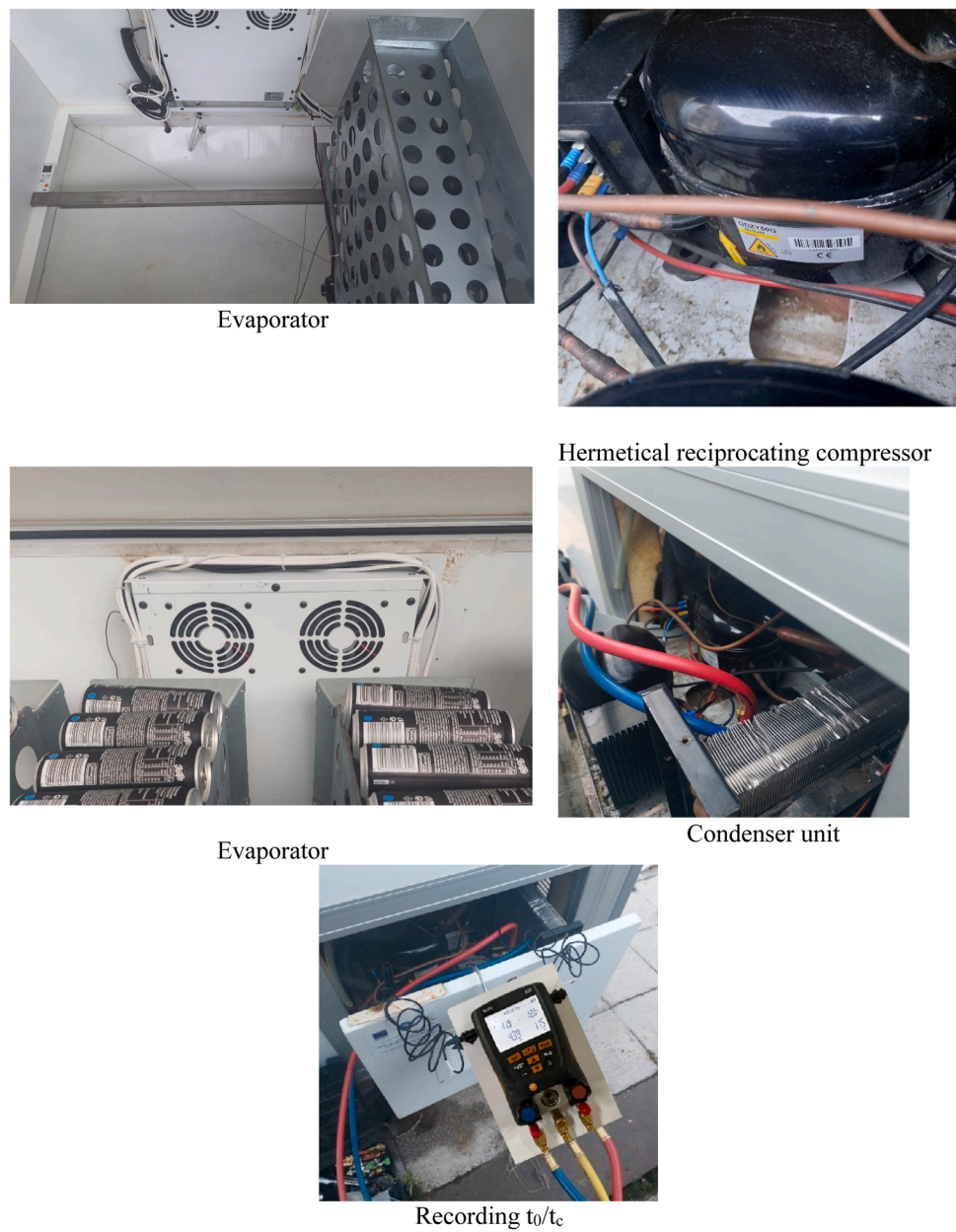


Fig. 16. Refrigeration unit main components.

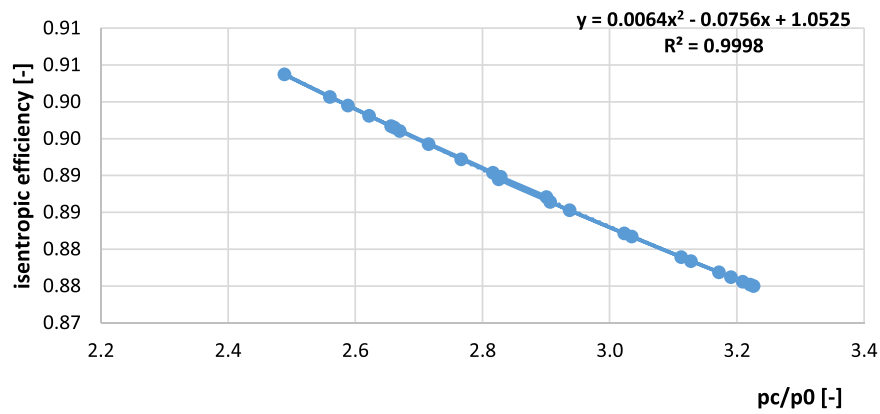


Fig. 17. Isentropic efficiency experimental value for hermetical reciprocating compressor R600a.

Jurges, Miheev, Mitchell, and Ashrae).

The experimental data demonstrates a clear diurnal variation, with power consumption peaking during midday and decreasing in the early morning and late evening, a pattern consistent with compressor behavior influenced by ambient temperature variations. The increase in power consumption during midday peaks is driven by higher heat loads into the vending machine due to increased ambient temperatures that require the compressor to work harder. The theoretical models generally follow the same trend as the experimental data, though there are slight deviations in magnitude at specific times. Among the correlations, the Nusselt Jurges model aligns most closely with the experimental data, particularly during the midday peak, while others, such as Miheev and McAdams, consistently underestimate power consumption during peak hours. Deviations are minimal during early morning hours but become more pronounced at peak operating times, suggesting that some correlations may not fully account for the impact of elevated ambient temperatures or other environmental factors. The increase in compressor power during midday emphasizes the need for effective cooling strategies, such as improved insulation or ventilation, to reduce energy consumption during peak heat periods. While all models capture the general behavior of the compressor's power consumption, the Nusselt Jurges correlation emerges as the most reliable under peak load conditions.

By calculating the hourly average of the Absolute Average Percentage Error (AAPE), the total energy consumption of the VCRS, as determined by the mathematical model compared to experimental data, shows an error below 5% for correlations 1, 2, and 4, while for correlations 3 and 5, the error ranges between 5% and 7%. The higher errors in some correlations, notably with the Miheev and McAdams correlations, indicate that these models may not fully account for external conditions such as airflow, that are influencing the heat transfer.

The results show good agreement, with a relative error of approximately 10% between the experimental and theoretical values (Fig. 19).

The specialized literature indicates that energy balance values in commercial refrigeration plants range between 10% and 20% [26].

The results suggest that the theoretical model accurately reflects the operational small-power refrigeration system. This level of error aligns with findings reported in similar studies. However, based on the graphical analysis of the error variations, and the findings stated above, the Nusselt Jurges correlation was selected and plotted in Fig. 20.

The graph presented in Fig. 20 demonstrates a significant correlation between the outdoor temperature and the compressor power values. As the outdoor temperature increases, there is a corresponding rise in compressor power, indicating a direct relationship between these variables. This alignment suggests that outdoor temperature fluctuations have a measurable impact on the energy consumption of the compressor, which is critical for optimizing system performance and energy efficiency. Such insights could be valuable for predictive maintenance and

operational adjustments in climate-dependent systems. The compressor power increases with increasing outdoor temperatures because the system needs to remove heat at a higher rate to maintain the interior at a constant temperature. The observed increase in compressor power with outdoor temperature aligns with thermodynamic expectations, as higher ambient temperatures elevate condensation pressure, thereby increasing compression work.

During the experiment, three distinct values for the evaporation temperature were recorded: 1.6 °C, 1.8 °C, and 2.6 °C. Figs. 21a, 21b and 21c. illustrate the variation of the global coefficient of performance ($COP_{globalcooling}$) as a function of the condensation temperature. The results indicate that under the operating conditions analyzed, the $COP_{globalcooling}$ decreases consistently with an increase in condensation temperature across all correlations studied in this research. The theoretical values of the global coefficient of performance (COP) for the VCRS, as calculated using the developed mathematical model, fall within the range of 1.292 to 1.476 for all the correlations considered. Notably, for the Nusselt Jurges correlation, the $COP_{globalcooling}$ values are slightly higher, ranging from 1.321 to 1.476. The observed trends align with theoretical expectations, confirming the model's applicability for evaluating and optimizing refrigeration systems under varying operational conditions.

A 1 K increase in outdoor temperature results in an average rise of 1.6% in the experimentally measured total power consumption. The decrease of COP with increased condensation temperature noticed in Fig. 20 is caused by the fact that less heat can be removed from the refrigeration system when its condensation temperature increases. The inverse correlation between global COP and condensation temperature is expected due to higher exergy losses at elevated condensation pressures, which decrease the refrigeration effect per unit work input. Based on the experimental data, the total power consumption was modeled using a simple regression analysis. The resulting expression, represented by Eq. (23), demonstrates a strong statistical correlation, with a coefficient of determination R^2 of 99.55%, indicating the accuracy of the regression model.

$$P_{Total} = 0.724 \cdot t_e + 26.473 \quad (23)$$

Fig. 22 illustrates the variation in total power consumption as a function of outdoor temperature, showing an increasing trend ranging from 41.61 W to 49.38 W for condensation temperatures between 30.9 °C and 42 °C. For every 1 K increase in condensation temperature, the total power consumption of the VCRS rises by 0.697 W. The increased total power consumption with the increased condensation temperature is because the compressor has to work harder when the temperature increases to remove a higher heat load.

On average, an older traditional vending machine consumes approximately 7 to 12 kWh of electricity per day, while a modern

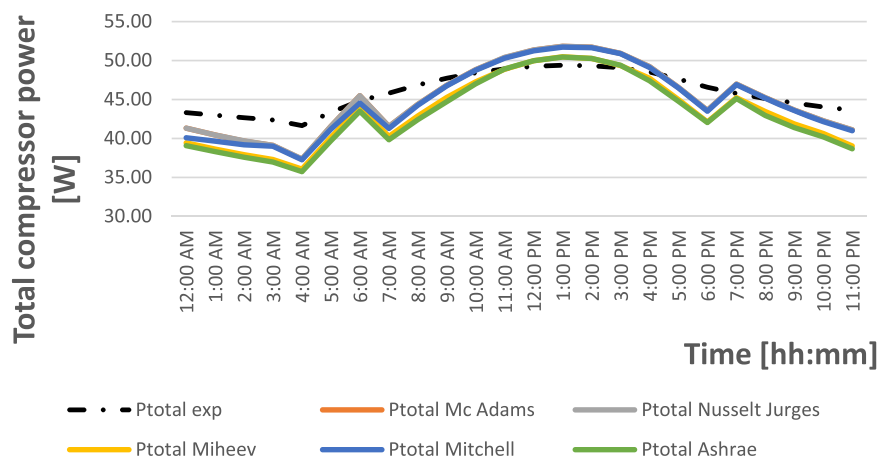


Fig. 18. Total power consumption - Experimental vs. Theoretical values.

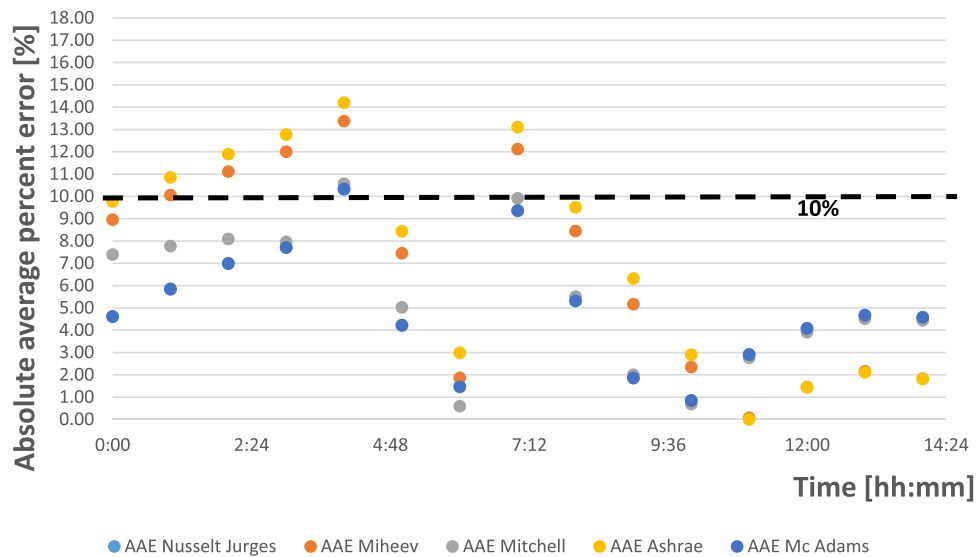


Fig. 19. Absolute average percent errors between theoretical data and experimental data.

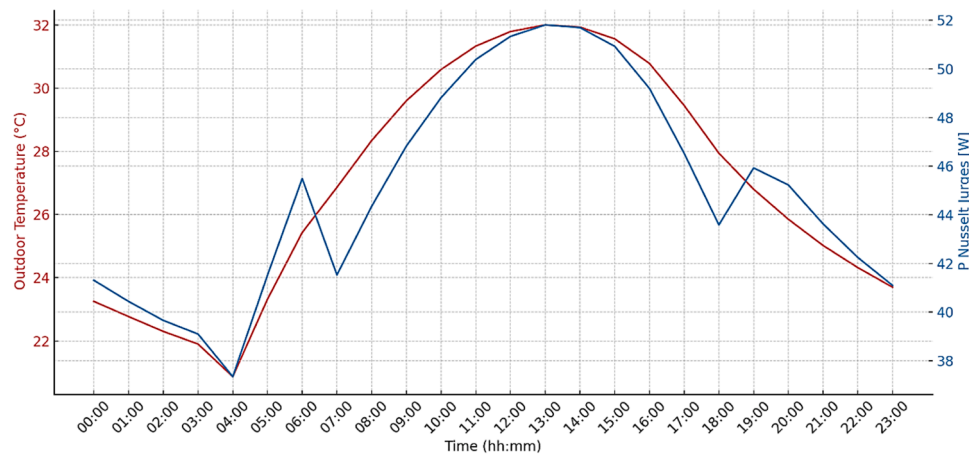


Fig. 20. Compressor power vs. outdoor temperature.

vending machine typically consumes around 5.7 kWh per day. The prototype that the authors developed operates off-grid using solar power, resulting in lower total power consumption. Experimental data indicates an energy consumption of 1.1 kWh per day; thus, an important energy economy is achieved.

The mathematical model can also be effectively applied to evaluate system performance with alternative refrigerants, making it a valuable tool for analyzing the potential effects of implementing a drop-in replacement process.

6. Conclusions

This study presents a significant advancement in the field of energy-efficient refrigeration systems through the development and validation of an innovative mathematical model specifically designed for a specific type of a vending machine. The SVIEE-R vending machine stands out from existing technologies by integrating a 12 V DC-powered refrigeration system using R-600a refrigerant, which significantly reduces energy consumption and CO₂ emissions. Unlike conventional vending machines, it operates independently of the electrical grid, relying on a solar-powered system with an advanced thermal insulation enclosure to enhance energy efficiency. Additionally, its automated charging and delivery system minimizes thermal losses, making it more sustainable

and cost-effective. The mathematical model that has been developed represents a valuable tool for evaluating refrigeration systems from multiple perspectives, including thermodynamic efficiency, geometric optimization, and overall performance.

By continuously monitoring the energy consumption of vending machines, opportunities for energy reduction can be identified, improving operational efficiency and reducing the environmental impact of Vapor Compression Refrigeration Systems (VCRS). Several studies have previously investigated the energy consumption of vending machines within university campuses across Europe [27–30], as well as in commercial and office buildings. This study brings the novelty of integrating the measured data into a mathematical modeling workflow that can be adjusted according to different inputs.

The objective of this research is to monitor the energy consumption of an off grid vending machine installed outdoors, utilizing R600a as the refrigerant in the VCRS. An eco-friendly vending machine will be designed to incorporate low-energy consumption technologies and will be powered by solar panels. This approach aims to minimize operating costs while significantly reducing energy consumption. The findings highlight that the evaporator's geometry significantly influences cooling capacity and the coefficient of performance (COP). Experimental validation of the model revealed that the COP, calculated using the Nusselt Jurges correlation, ranged from 1.321 to 1.476, demonstrating high

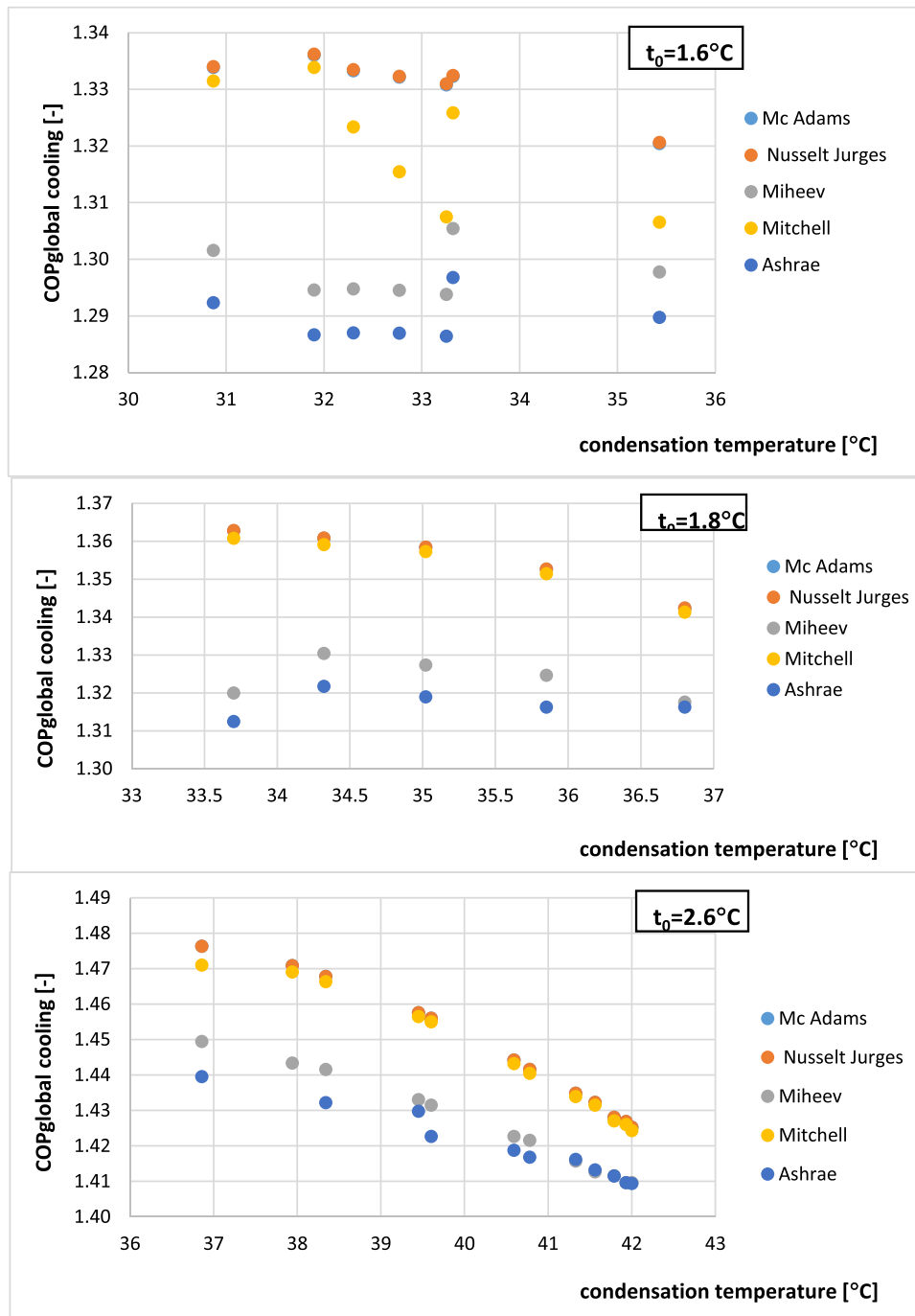


Fig. 21. Variation of $COP_{global_cooling}$ with condensation temperature at evaporation temperatures between 1.6...2.6 °C.

energy efficiency under varying operating conditions. The analysis also confirmed that for every 1 K increase in outdoor temperature, total power consumption increased by approximately 1.6%, with condensation temperatures ranging from 30.9 °C to 42 °C causing the power consumption to rise between 41.61 W and 49.38 W. This finding underlines the importance of optimizing geometric and thermodynamic parameters to achieve energy-efficient performance.

Using the mathematical model, deviations between theoretical predictions and experimental data were kept below 10%, with correlations such as Nusselt Jurges achieving the highest accuracy. The integration of R600a refrigerant, characterized by its negligible Global Warming Potential (GWP = 3) and zero Ozone Depletion Potential (ODP), aligns with sustainability goals. This refrigerant, combined with innovative

design features, such as the advanced thermal insulation system with a total thermal conductivity of approximately 0.05 W/(m·K), positions the SVIEE-R vending machine as a state-of-the-art solution for energy-efficient refrigeration.

The system's energy independence is achieved through a photovoltaic array with a maximum power output of 405 W, managed by an MPPT controller with a DC/DC conversion efficiency of 98%. Data collected from the HIOKI PW3198 analyzer showed high agreement between monitored and theoretical power consumption values, confirming the reliability of the model.

The innovation and utility of this vending machine lie not only in its energy efficiency but also in its design adaptability for off-grid locations and urban environments. Compared to conventional vending machines,

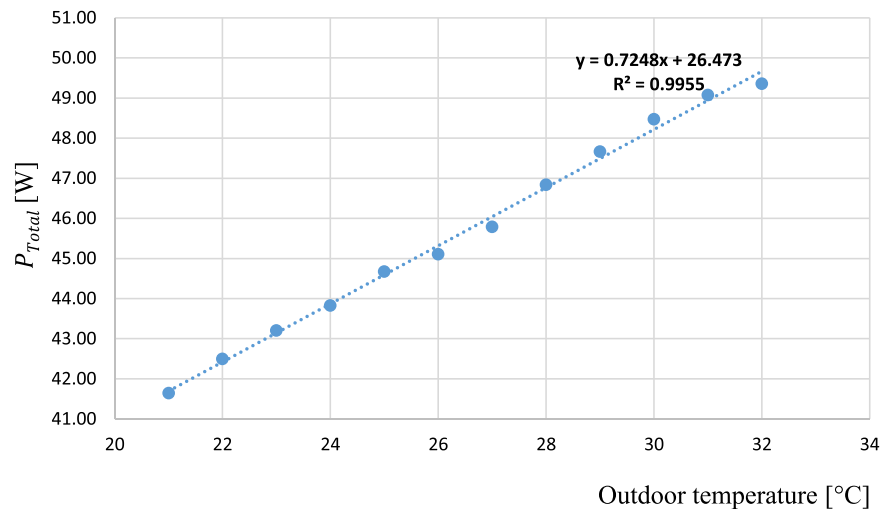


Fig. 22. Total power consumption experimental vs. outdoor temperature.

the SVIEE-R delivers superior performance with a validated reduction in energy consumption and carbon footprint. Additionally, it offers improved operational flexibility through features like QR-code-enabled product delivery and a fully automated control system. The findings provide valuable insights into how R-600a and advanced modeling techniques improve energy performance while reducing environmental impact. This research offers a practical framework for applying theoretical modeling to real-world small-scale refrigeration systems in sustainable engineering.

The research on the concept of a vending machine featuring a thermally insulated tank for horizontally stored products, with a top-mounted power supply and a refrigeration system powered by 12 V direct current, has presented significant technical challenges. The tests conducted and the results presented in this article open opportunities not only for the development of an industry dedicated to manufacturing vending machines but also for domestic applications in regions where a mains supply for a refrigerator is unavailable.

Future improvements will focus on refining the model to account for dynamic scenarios, including fluctuating user demand. Moreover, alternative refrigerants with lower environmental impact will be explored, alongside advanced control algorithms for predictive maintenance and further energy optimization.

CRediT authorship contribution statement

Alina Girip: Validation, Supervision, Methodology, Data curation, Conceptualization. **Răzvan Calotă:** Writing – original draft, Project administration, Investigation, Funding acquisition, Conceptualization. **Ioan Mihail Savaniu:** Writing – original draft, Resources, Project administration, Data curation. **Anica Ilie:** Supervision, Investigation, Formal analysis. **Emanuil Ovadiuc:** Investigation. **Oana Tonciu:** Visualization, Methodology, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Savaniu Ioan Mihail has patent SISTEM DE VANZARE AUTOMAT INDEPENDENT ENERGETIC A PRODUSELOR ALIMENTARE - VENDING SYSTEM ENERGY INDEPENDENT FOR FOOD pending to A100779. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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